



Exercise Sheet 07

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Total points: 12

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a jupyter notebook.

1. Hypothesis testing

- (a) For this exercise, we will analyze empirical data from students taking exams. You can download the dataset from the following link: [Student data](#). The dataset contains marks obtained by students in basic skills such as math and language (reading and writing).

Let's assume this data has been collected from an educational institution, and we have been tasked with drawing important inferences to simulate a real-world scenario. To achieve this, we have formulated hypotheses about the population and will test them using statistical tests.

Hint: Use the following code for plotting the distribution and the lines:

```
from scipy.stats import t  
t.pdf and t.ppf.
```

- i) Open the dataset and form the "Overall Performance" column, which is the average of all three scores of a student. Here is the Python code to read the downloaded dataset using pandas:

```
1 import pandas as pd  
2  
3 df = pd.read_csv('StudentsPerformance.csv')
```

- ii) Conduct a one-sample t-test (one-sided) under the following conditions. Additionally, plot the distribution, critical value, and t-test score. Do we reject the null hypothesis?

Upon examination of the data, we discover that the mean overall performance of the students in our sample is 67.77. Now, the question arises: is this variation simply a result of random chance, or does it accurately represent the larger population? To determine this, we will employ a hypothesis test and draw meaningful inferences.

The null and alternative hypotheses are defined as follows:

H_0 : The overall performance of students is greater than or equal to a respectable score of 70.

H_a : The overall performance of students is less than 70.

Since Type 2 error is more consequential, we will increase the significance level to 0.10 from the standard 0.05, as we want to be cautious about failing to reject H_0 when it is false.

- iii) Perform a two-sample t-test (Independent t-test) under the following conditions. Also, plot the distribution, upper and lower critical values, and the t-test score. Do we reject the null hypothesis?

4P

H_0 : There is no difference in overall performance of students between those who have taken test preparation and those who have not.

H_a : There is a difference in overall performance of students between those who have taken test preparation and those who have not.

```
1 print('Mean score of students with test preparation:', df['  
    Overall Performance'][df['test preparation course'] == '  
    completed'].mean())  
2 print('Mean score of students without test preparation:', df['  
    Overall Performance'][df['test preparation course'] == 'none'  
    ].mean())
```

```
1 >>> Mean score of students with test preparation:  
    72.66945996275605  
2 >>> Mean score of students without test preparation:  
    65.03894080996888
```

We observe that students who have taken test preparation have achieved better scores compared to their peers. We need to validate this observation.

Since Type 2 error is more consequential, we will increase the significance level to 0.10 from the standard 0.05, as we want to be cautious about failing to reject H_0 when it is false.

- (b) When developing a predictive model, selecting influential predictor variables is crucial. While trying all possible variable combinations for the optimal model is time-consuming, an alternative method is needed to obtain a decent model.

2P

In linear regression, the p-value method is used to assess the significance of predictor variables. A low p-value indicates that a variable has a significant impact on the model's outcome, while a high p-value suggests insignificance. This method helps in selecting important features for the model.

Consider the Housing dataset, containing property prices in the Delhi region. Our objective is to optimize property sale prices by considering factors like area, bedrooms, parking, etc.

- i) Numerical data is preferred in machine learning models because algorithms are designed to work with numbers efficiently, enabling mathematical operations, distance calculations, and easy feature engineering. Non-numerical data can be encoded. Use the following piece of code to encode non-numerical columns.

```
1 df = pd.read_csv('Housing.csv')  
2  
3 from sklearn.preprocessing import LabelEncoder  
4 # encode the numerical values  
5 df.iloc[:, :-1] = df.iloc[:, :-1].apply(LabelEncoder().  
    fit_transform)  
6 # get one-hot encoding of the last column  
7 one_hot = pd.get_dummies(df['furnishingstatus'])  
8 # drop the last column and substitute it with the columns from one  
    -hot  
9 df = df.drop(['furnishingstatus'], axis=1)  
10 df = pd.concat([df, one_hot], axis=1)
```

- ii) Fit a regression line to the data using OLS from `statsmodels`. Identify the variables that are not significant at a 5% significance level.

2. Monty Hall problem

- There are three doors in front of you, two of them are hiding goats, and the third one is hiding a car, but you do not know where is the car.
- In the first step of the game, you choose a door. This door remains closed, and you do not know what is behind it.
- In the second step of the game, the host opens one of the remaining two doors. The host, which knows where the goats are and where the car is, always picks a door which hides a goat.
- Then, the host offers you a chance to change your choice of the door. If you decline, you will get what is behind the door you initially chose, if you accept, you will get what is behind the remaining door, which was neither initially chosen by you, nor opened by the host in the second step.
- Would you change your choice?



Drawing credits: stable diffusion and Luka Petrović

- (a) Would you change your initial choice? What are the odds to win the car?
- (b) Answer the above question with the help of a causal diagram.

1P

1P

3. Distribution of estimators in linear regression

Consider 1000 random samples of $n = 100$ realizations of bivariate RV (X, Y) where $X \sim \text{Norm}(0,10)$ and $Y \sim \text{Norm}(0,10)$ are independent, i.e. there is no linear relationship.

- (a) For each sample of $n = 10$ realizations infer the intercept parameter $\hat{\beta}_0$ and slope parameter $\hat{\beta}_1$ for the linear regression. (You can use the notebook 11.03_linear_relations for this task)
- (b) For each fitted line calculate the standard error to estimate the standard deviation σ of fitted parameter $\hat{\beta}_1$ from the true mean $\beta_1 = 0$.
- (c) For a normal distribution, 95% of the samples lie in the interval $[-2\sigma + \mu, \mu + 2\sigma]$ where μ is the mean. Therefore, the intervals $[-2\sigma + \hat{\beta}_1, \hat{\beta}_1 + 2\sigma]$ should contain 0 in 95% of all cases. Can you confirm this?

1P

2P

1P